



TEKATHON-5.0- 2026



Problem Statement ID – 26168

AI-ML based Intelligent Dead Reckoning system for seamless navigation

Theme – Miscellaneous • PS Category – Software

Organization – ISRO

NavDR | Neural-Augmented DR



IDEA TITLE

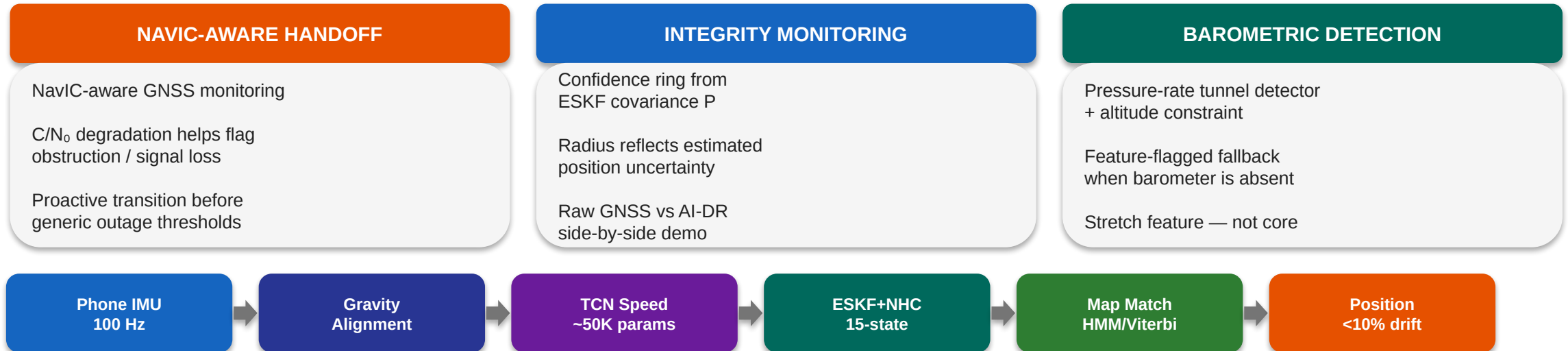
❖ NavDR — AI-ML Intelligent Dead Reckoning for Seamless Navigation

PROBLEM

- GNSS fails in tunnels, parking garages and urban canyons → navigation freezes
- Pure IMU integration accumulates drift rapidly

SOLUTION

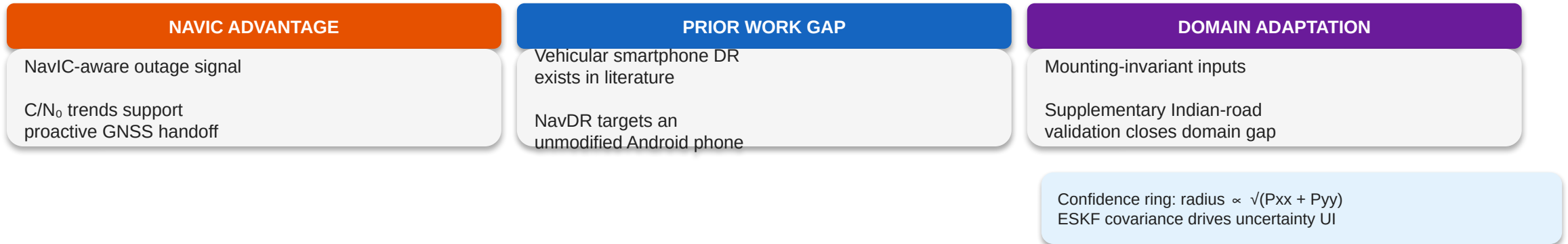
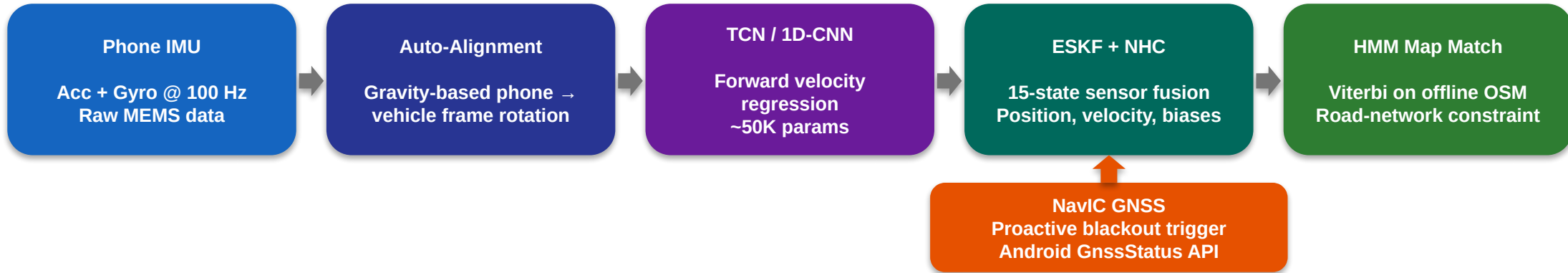
- TCN learns forward velocity from phone IMU → ESKF fuses motion → HMM constrains to OSM roads
- **Target: <10% drift during GNSS denial • IO-VNBD benchmark + Indian-road validation**





TECHNICAL APPROACH

- Python • PyTorch/ONNX Runtime • Android Kotlin • OSMDroid • Valhalla Meili • IO-VNBD





FEASIBILITY AND VIABILITY

- Real risks, concrete mitigations

RISK		MITIGATION
Android Doze mode can reduce IMU sampling	➔	Foreground Service + partial WakeLock; verify sustained 100 Hz
Phone shifts / rotates on mount	➔	Monitor alignment; recompute phone → vehicle rotation
IO-VNBD roads differ from India	➔	Indian test-drive validation + augmentation / transfer learning
GNSS multipath gives a wrong but confident fix	➔	Innovation / chi-square gating rejects inconsistent updates
ESKF math is the critical-path risk	➔	Adapt open-source ESKF; fallback to simpler 2D KF



IMPACT AND BENEFITS

- **Smartphone-first dead reckoning — zero dedicated vehicle hardware**

SAFETY

Continuous navigation through tunnels & underpasses

Emergency vehicles retain route awareness

STRATEGIC

NavIC-aware GNSS handoff

Builds resilience around India's own navigation system

ECONOMIC

₹0 dedicated hardware

Scales through software deployment

SOCIAL

Budget-phone friendly

Useful for delivery, taxi, logistics and personal nav

Target: <10% positional drift

On-device inference: low latency

No OBD-II / dedicated hardware

USE CASES: highway tunnels • multi-level parking • urban canyons • forest roads • GNSS-denied zones

USERS: emergency responders • logistics fleets • ride-hailing • delivery • personal navigation

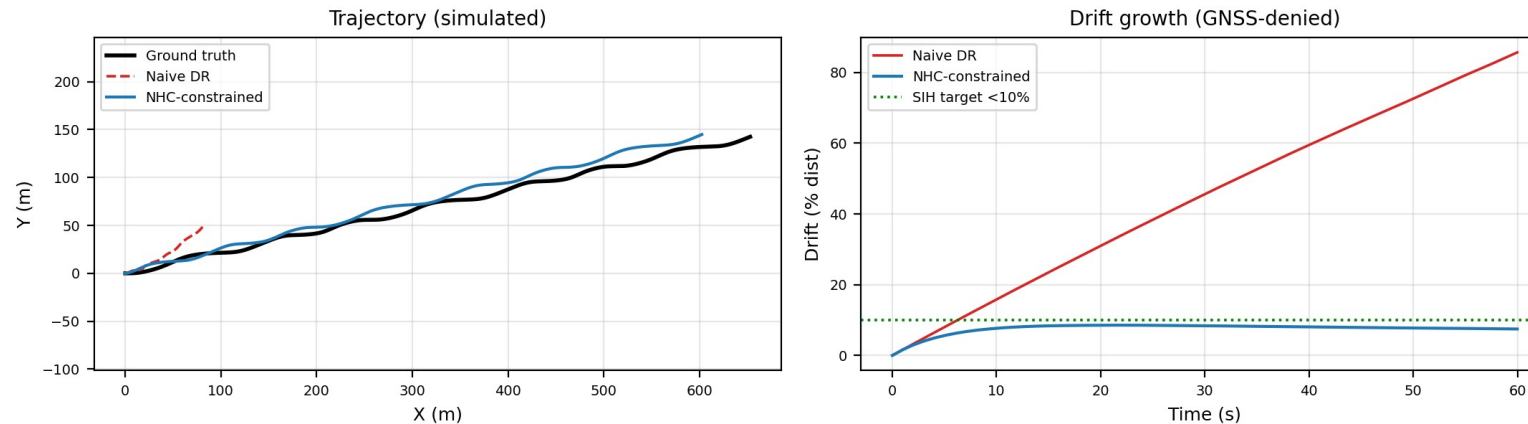


RESEARCH AND REFERENCES

- **Research basis and reference work**

- [1] Onyekpe et al. — IO-VNBD benchmark dataset. Data in Brief (2021). DOI: 10.1016/j.dib.2021.106885
- [2] Brossard et al. — AI-IMU Dead-Reckoning. IEEE T-IV (2020). DOI: 10.1109/TIV.2020.2980758
- [3] Sirikonda et al. — Tightly Coupled NavIC + Low-Cost Sensors. IEEE Sensors J. (2024). DOI: 10.1109/JSEN.2024.3367927
- [4] Jayanth et al. — EqNIO: Subequivariant Neural Inertial Odometry. ICLR (2025).
- [5] Onyekpe et al. — R-WhONet: Transfer Learning for Vehicular Positioning. NCA (2025). DOI: 10.1007/s00521-024-10046-z
- [6] Valhalla — Meili HMM Map Matching engine. Open-source implementation.
- [7] Android GnmssStatus / raw GNSS APIs — on-device GNSS signal monitoring.
- [8] Groves / Solà — integrated navigation and error-state Kalman filter foundations.

Preliminary result (simulated baseline — to be replaced with an IO-VNBD-trained result per PS requirement):



Naive strapdown DR: 85.7% drift over 60s | NHC-constrained (proxy): 7.5% drift — meets SIH <10% target